41.1: An animal's diet must supply chemical energy, organic building blocks, and essential nutrients

Although dining on sea stars, crabs, and fish is the herring gull's specialty, all animals eat other organisms—dead or alive, piecemeal or whole. Unlike plants, animals must consume food for both energy and the organic molecules used to assemble new molecules, cells, and tissues. Overall, an adequate diet must satisfy three needs: chemical energy for cellular processes, organic building blocks for macromolecules, and essential nutrients. The process by which an animal takes in and makes use of food to meet these needs constitutes **nutrition**.

All of an animal's activities depend on sources of chemical energy in the diet: carbohydrates, proteins, and lipids. The chemical reactions of cellular respiration use this energy to produce ATP, which powers processes ranging from DNA replication and cell division to vision and flight (see Concept 8.3). Ingested and digested nutrients can be broken down immediately or stored for future use.

In addition to fuel for ATP production, an animal requires raw materials for biosynthesis. To build the complex molecules it needs to grow, maintain itself, and reproduce, an animal's food must provide a source of organic carbon (such as sugar) and a source of organic nitrogen (such as protein).

The third requirement of an animal's diet is to provide **essential nutrients**, substances that an animal requires but cannot assemble from simple organic molecules.

Essential Nutrients

Essential nutrients in the diet include certain amino acids and fatty acids, as well as vitamins and minerals. The key functions of essential nutrients include serving as substrates of enzymes, as coenzymes, and as cofactors in biosynthetic reactions **(Figure 41.2)**.

Figure 41.2

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Roles of essential nutrients.
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The enzyme fatty acid desaturase catalyzes the reaction that converts linoleic acid to gamma-linoleic acid, a precursor for other cellular components. This reaction showcases the four classes of essential nutrients.

The enzyme fatty acid desaturase catalyzes the reaction that converts linoleic acid (an essential fatty acid with a hydrocarbon tail that is the substrate in this reaction) to gamma-linoleic acid. Gamma-linoleic acid is a precursor used in phospholipids (cell membrane components) and prostaglandins (used in cell signaling). This reaction showcases the four classes of essential nutrients, which are, 1, essential fatty acids; 2, vitamins (coenzyme); 3, minerals (cofactor); and 4, essential amino acids (monomers of polypeptides). The enzyme fatty acid desaturase contains the essential nutrients: vitamin B 3 (coenzyme), the mineral iron (cofactor), and a polypeptide chain consisting in parts of essential amino acids (monomers of polypeptides) including tyrosine, phenylalanine, leucine, and isoleucine.

This example of a biosynthetic reaction illustrates some common functions for essential nutrients. The conversion of linoleic acid to γ-linoleic acid by the enzyme fatty acid desaturase involves all four classes of essential nutrients, as labeled in blue. Note that almost all enzymes and other proteins in animals contain some essential amino acids, as indicated in the partial sequence shown for fatty acid desaturase.

In general, an animal can obtain all essential amino acids and fatty acids, as well as vitamins and minerals, by feeding on plants or other animals. Needs for particular nutrients vary among species. For instance, some animals (including humans) must get ascorbic acid (vitamin C) from their diet, whereas most animals can synthesize it from other nutrients.

Essential Amino Acids

All organisms require a standard set of 20 amino acids to make a complete set of proteins (see Figure 5.14). Plants and microorganisms normally can produce all 20. Most animals have the enzymes to synthesize about half of these amino acids, as long as their diet includes sulfur and organic nitrogen. The remaining amino acids must be obtained from the animal's food in prefabricated form and are therefore called **essential amino acids**. Many animals, including adult humans, require eight amino acids in their diet: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. (Human infants also need a ninth, histidine.)

The proteins in animal products such as meat, eggs, and cheese are "complete," providing all the essential amino acids. In contrast, most plant proteins are "incomplete," being deficient in one or more essential amino acids. Corn (maize), for example, is deficient in tryptophan and lysine, whereas beans are lacking in methionine. However, vegetarians can easily obtain all of the essential amino acids by eating a varied diet of plant proteins.

Essential Fatty Acids

Animals require fatty acids to synthesize a variety of cellular components, including membrane phospholipids, signaling molecules, and storage fats. Although animals can synthesize many fatty acids, they lack the enzymes to form the double bonds found in certain required fatty acids. Instead, these molecules must be obtained from the diet and are considered **essential fatty acids**. In mammals, they include linoleic acid (see Figure 41.2). Animals typically obtain ample quantities of essential fatty acids from seeds, grains, and vegetables in their diet.

Vitamins

As Albert Szent-Györgyi, the discoverer of vitamin C, once quipped, "A vitamin is a substance that makes you ill if you *don't* eat it." **Vitamins** are organic molecules that are required in the diet in very small amounts (0.01–100 mg per day, depending on the vitamin).

The 13 vitamins required by humans vary in both chemical properties and function (**Table 41.1**). Vitamin B2, for example, is a water-soluble vitamin that is converted in the body to FAD, a coenzyme used in many metabolic processes, including cellular respiration (see Figure 9.12). Vitamin C, which is required for the production of connective tissue, is also water-soluble.

Table 41.1

Vitamin Requirements of Humans

A table lists the vitamin requirements of humans, including the major dietary sources, functions in the body, and symptoms of deficiency.

The table is divided into two sections as follows. Water soluble vitamins and fat-soluble vitamins.

Table 41.1 Vitamin requirements of Humans
Vitamin
Water-soluble vitamins
B 1, thiamine
B 2, riboflavin
B 3, niacin
B 5, pantothenic

B 6, pyridoxine
B 7, biotin
B 9, folic acid
B 12, cobalamin
C, ascorbic acid
Fat-soluble vitamins
A, retinol
D
E, tocopherol
K, phylloquinone

Fat-soluble vitamins include vitamin A, which is incorporated into visual pigments of the eye, and vitamin D, which aids in calcium absorption and bone formation. Our dietary requirement for vitamin D, unlike other vitamins, turns out to be variable. Why? When our skin is exposed to sunlight, our bodies synthesize vitamin D, reducing our dietary need.

For people with imbalanced diets, taking vitamin supplements at recommended daily levels is reasonable. It is far less clear that massive doses of vitamins confer any health benefits or are even safe. Moderate overdoses of water-soluble vitamins are probably harmless because excesses are excreted in urine. However, excesses of fat-soluble vitamins are deposited in body fat, so overconsumption may cause them to accumulate to toxic levels.

Minerals

Dietary **minerals** are inorganic nutrients, such as iron and sulfur, that are usually required in small amounts—from less than 1mg to about 2,500mg per day. As shown in **Table 41.2**, minerals have diverse functions in animal physiology. Some are assembled into the structure of proteins; iron, for example, is incorporated into the oxygen carrier hemoglobin as well as some enzymes (see Figure 41.2). Others, such as sodium, potassium, and chloride, are important in the functioning of nerves and muscles and in maintaining osmotic balance between cells and the surrounding body fluid. In vertebrates, the mineral iodine is incorporated into thyroid hormone, which regulates metabolic rate. Vertebrates also require relatively large quantities of calcium and phosphorus for building and maintaining bone.

Table 41.2

Mineral Requirements of Humans*

A table lists the mineral requirements of humans, including the major dietary sources, functions in the body, and symptoms of deficiency.

The table is as follows.

Table 41.2 Mineral requirements of Humans asterisk
Mineral
Calcium, C a
Phosphorous, P
Sulfur, S
Potassium, K
Chlorine, C I
Sodium, N a
Magnesium, M g
Iron, F e
Fluorine, F
Iodine, I
Asterisk Additional minerals required in trace amounts include Cobalt (C o), Copper (C u), manganese (M n), molybdenum (M o), Selenium (S e), and zinc (Z n). All of these minerals, as well as those in the table, can be harmful in excess. Note: following minerals required more than 200 milligrams per day: calcium, phosphorous, sulfur, potassium, chlorine, sodium, and magnesium.

Ingesting too much of some minerals can cause health problems. For example, excess sodium can contribute to high blood pressure. This is a particular problem in the United States, where the typical person consumes enough salt (sodium chloride) to provide about 20 times the required amount of sodium. Processed foods often contain large amounts of sodium chloride, even if they do not taste salty.

Variation in Diet

Despite many shared nutritional needs, animals have diverse diets. **Herbivores**, such as cattle, sea slugs, and caterpillars, dine mainly on plants or algae. **Carnivores**, such as sea otters, hawks, and spiders, mostly eat other animals. Bears, crows, cockroaches, and other **omnivores** (from the Latin *omnis*, all) don't in fact eat everything, but they do regularly consume animals as well as plants or algae. Humans are typically omnivores or herbivores.

The terms *herbivore, carnivore*, and *omnivore* represent the kinds of food an animal usually eats. However, most animals are opportunistic feeders, broadening their diet when their usual foods aren't available. For example, deer are herbivores, but occasionally eat insects, worms, or bird eggs. Similarly, herring gulls eat marine invertebrates, insects, and small fishes, but also human refuse. Note that microorganisms are an unavoidable "supplement" in every animal's diet.

Dietary Deficiencies

A diet that lacks one or more essential nutrients or consistently supplies less chemical energy than the body requires results in *malnutrition*, a failure to obtain adequate nutrition. Malnutrition affects one out of four children worldwide, impairing health and often survival.

Deficiencies in Essential Nutrients

Insufficient intake of essential nutrients can cause deformities, disease, and even death. For example, deer and other herbivores can develop fragile bones if the plants they consume grew in phosphorus-deficient soil. In such environments, some grazing animals obtain missing nutrients by consuming concentrated sources of salt or other minerals **(Figure 41.3)**. Similarly, some birds supplement their diet with snail shells, and certain tortoises obtain minerals from stones they ingest.

Figure 41.3

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Obtaining essential nutrients from an unusual source.
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Chamois kneels to lick salt.

A juvenile chamois (Rupicapra rupicapra), an herbivore, licks salts from exposed rocks in its alpine habitat. This behavior is common among herbivores that live where soils and plants provide insufficient amounts of minerals.

Like other animals, humans sometimes have diets lacking in essential nutrients. A diet with insufficient amounts of one or more essential amino acids causes protein deficiency, the most common type of

malnutrition among humans. In children, protein deficiency may arise if their diet shifts entirely from breast milk to foods that contain relatively little protein, such as rice. Such children, if they survive infancy, often have impaired physical and mental development.

Undernourishment

As mentioned earlier, malnutrition can also be caused by a diet that fails to provide enough chemical energy. In this situation, the body first uses up stored carbohydrates and fat. It then begins breaking down its own proteins for fuel: Muscles shrink, and the brain may become protein-deficient. If energy intake remains less than energy expenditures, the animal will eventually die. Even if a seriously undernourished animal survives, some of the damage may be irreversible.

Inadequate nourishment in humans is most common when drought, war, or other crisis severely disrupts the food supply. The United Nations reported in 2021 that more than 34 million people globally were facing acute hunger, including many people in Yemen, South Sudan, northern Nigeria, Ethiopia, and Madagascar. In these and many other nations, conflict and severe climate events restricting the food supply have been further compounded by the COVID-19 pandemic.

Sometimes undernourishment occurs within well-fed human populations as a result of eating disorders. For example, anorexia nervosa involves weight loss to a level that is unhealthy for the individual's age and height and may be related to a distorted body image.

Assessing Nutritional Needs

Determining the ideal diet for the human population is an important but difficult problem for scientists. As objects of study, people present many challenges. Unlike laboratory animals, humans are genetically diverse. They also live in settings far more varied than the stable and uniform environment that scientists use in laboratory experiments. Ethical concerns present an additional barrier. For example, it is not acceptable to investigate the nutritional needs of children in a way that might harm a child's growth or development.

Many insights into human nutrition have come from *epidemiology*, the study of human health and disease at the population level. In the 1970s, for instance, researchers discovered that children born to women of low socioeconomic status were more likely to have neural tube defects, which occur when tissue fails to enclose the developing brain and spinal cord (see Concept 47.2). The English scientist Richard Smithells thought that malnutrition among these women might be responsible. As described in **Figure 41.4**, he found that vitamin supplementation greatly reduced the risk of neural tube defects. In other studies, he obtained evidence that folic acid (vitamin B9) was the specific vitamin responsible, a finding confirmed by other researchers. Based on this evidence, the United States in 1998 began to require that folic acid be added to

enriched grain products used to make bread, cereals, and other foods. Follow-up studies have documented the effectiveness of this program in reducing the frequency of neural tube defects. Thus, at a time when microsurgery and sophisticated diagnostic imaging dominate the headlines, a simple dietary change such as folic acid supplementation may be among the greatest contributors to human health.

Figure 41.4

Inquiry: Can diet influence the frequency of neural tube defects?

Experiment

Richard Smithells, of the University of Leeds, in England, examined the effect of vitamin supplementation on the risk of neural tube defects. People who had given birth to one or more babies with such a condition were put into two study groups. The experimental group consisted of those who were planning a pregnancy and began taking a multivitamin at least four weeks before attempting conception. The control group, who were not given vitamins, included those individuals who declined them and those who were already pregnant. The numbers of neural tube defects resulting from the pregnancies were recorded for each group.

Results

A table illustrates the vitamin supplements experimental and control groups for number of infants/fetuses studied and infants/ fetuses with aneural tube defect.

Data from R. W. Smithells et al., Possible prevention of neural-tube defects by periconceptional vitamin supplementation, *Lancet* 315:339–340 (1980).

Group	Number of Infants/Fetuses Studied	Infants/Fetuses with a Neural Tube Defect
Vitamin supplements (experimental group)	141	1
No vitamin supplements (control group)	204	12

Conclusion

This controlled study provided evidence that vitamin supplementation protects against neural tube defects, at least after the first pregnancy. Follow-up trials demonstrated that folic acid alone provided an equivalent protective effect.

INTERPRET THE DATA *After folic acid supplementation became standard in the United States, the frequency of neural tube defects dropped to an average of just one in* 5,000 *live births. Propose two explanations of why the observed frequency was much higher in the experimental group of the Smithells study.*

INQUIRY IN ACTION Read and analyze the original paper in *Inquiry in Action: Interpreting Scientific Papers*.